

Paper Robust Statistics

Luisa Scaroni r1085999

June 2026

1 Introduction

The goal of the project is to perform a robust regression analysis where the observations may contain rowwise and cellwise outliers. We extracted a random subset of 500 observations from the US Cancer dataset compiled by the National Cancer Institute for the period 2010-2016. The original dataset contained 3047 US counties and 27 numerical explanatory variables covering socio-economic and health indicators. Subsequently, through the DDC method we detected cellwise outliers and then fitted a robust regression model for the variable `targetdeathrate` via an MM-estimator.

2 Exploratory Data Analysis

The exploratory data analysis began with the *summary()* function, revealing that the variables operate on vastly different scales. Then, we studied the data distribution since the DDC algorithm assumes variables **without heavy skewness**. Firstly, we noticed that the majority of the variables are heavily right skewed, such as `medianage` with a coefficient of 10.03. Meanwhile, only `pctwhite` is strongly left-skewed (-1.51).

Furthermore, we computed the robust correlation matrix to check the relations among variables, noting that the robust approach yields more reliable estimates compared to the classical one, which is extremely sensible to outliers. The difference in values between the classical and the robust corplot is clear evidence that the outliers are deflating classical dependencies. In addition, the target variable is positively correlated with socioeconomic

distress indicators like `povertypercent` and `pctunemployed16_over`. Conversely, it is negative correlated with median incomes and higher education levels. The matrices also revealed significant **multicollinearity** among several variables, for example an inverse relationship between `medincome` and `povertypercent`.

Next, we plotted histograms and QQ-plots to study normality. The inspection of the histograms showed that, as assumed before, the majority of the variables are heavily right-skewed, indeed the data mass is concentrated at lower values. The QQ-plots exhibited deviations from normality, with only a few exceptions like `pctpubliccoverage` where both the QQ-plot and the histogram appear normal. We also employed the **Shapiro-Wilk normality test**, the results rejected the null hypothesis of normality ($p < 0.05$) for all variables except `pctpubliccoverage`, `pctempprivcoverage` and `medianagefemale`. A previous assessment confirmed the presence of **no missing values**.

To have an initial view on outliers, we decided to employ adjusted boxplots. This is the best choice since the classic boxplots would misclassify many tail observations as anomalous due to the severe right-skewed behavior. Some of the variables clearly contain outliers, an example is `medianage` with values larger than 400. To mitigate the skewness, a **data transformation** was necessary. At first, our idea was to use the Box-Cox transformation, but after noticing that some variables did not contain strictly positive values, we discarded this option. We could have overcome this problem by adding a small constant like 0.01 to the variables, but we decided to proceed differently to avoid an artificial shift in the distribution and a distortion of the data structure. We concluded that the most suitable approach was to apply the robust **Yeo-Johnson transformation**:

```
1 transform=transfo(xmatrix, type='YJ', robust=TRUE)
```

This method provided a good solution. We were able to achieve a skewness between -1 and 1 for almost all variable except `mediange`, which contains data entry errors. However, despite the improvement in the Shapiro-Wilk test value, we could not achieve perfectly Gaussian distributions. It's quite important to notice that the DDC algorithm works optimally under Normal assumption, but it is not a fundamental prerequisite like the absence of heavy skewness.

3 Cellwise and Rowwise

Due to the high dimensionality of the entire transformed dataset we decided to visualize only fifteen counties. To select them, we detected the **anomalies** by calculating the sum of the absolute standardized residuals for each row and then we chose the top fifteen counties with the highest values.

Before analyzing the *cellMap*, it is helpful to introduce the color-coding scheme: **yellow** indicates that the observed values aligns with the algorithm's prediction, **red** implies a value significantly higher than expected and **blue** signifies a value significantly lower than expected. We can observe that none

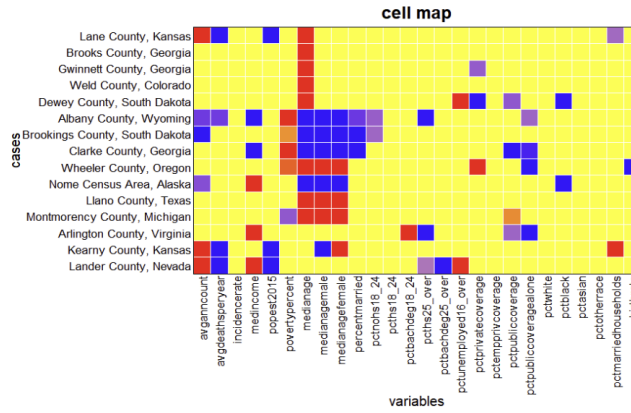
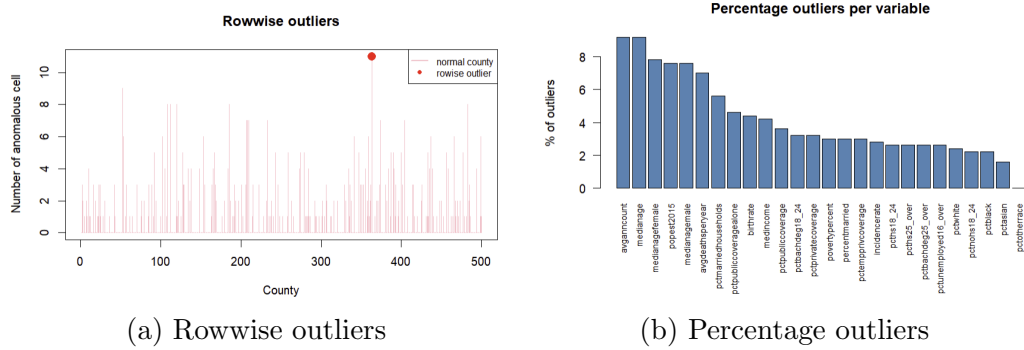


Figure 1: Outliers Map through DDC algorithm (CellMap)

of the rows are entirely blue

at Kearny County, Lander County and Lane County, we found an extremely high average number of cancer cases but an extremely low mortality rate (around 0.2%). We can assume that the data in `avgnncount` for these rows are probably wrong, indeed public records show that Lane county has a population of approximately 2,000 residents, consequently an average of 1,962 new cancer cases per year is impossible. Conversely, other extreme values are connected to socio-economic disparities. Arlington County, for example, is one of the wealthiest counties in the US, explaining the high `medincome`. In contrast, Clarke County is a lower-income area.

Next, we numerically evaluated the rowwise outliers. The algorithm identified merely a single observation as a **rowwise outlier**, while overall 4.146% of the total cells were classified as anomalous. The bar plot shows the distribution of cellwise outliers across each covariate. We can see that `medianage` and `avgnncount` are the most heavily contaminated ones likely because of the data entry error. Meanwhile `pctotherrace` is the least contaminated variable. We generated a plot to identify row wise anomalies. The majority of the vertical lines are short, implying that most counties contain only about 2 anomalous cells. However, a red dot stands out, corresponding to a county with more than 10 anomalous cells. This likely corresponds to the Albany County which exhibits the highest number of non yellow cells in Figure 1 and also matched the index number 363.



4 Regression Analysis

After importing `cellLTS_code.R`, we studied the influence of outliers on the regression models. We defined the target variable (`target_deathrate`) and

constructed two dataset: **dataraw**, which uses the YJ transformed but contaminated matrix (**xtrans**), and **dataddc**, which uses the transformed and imputed matrix (**ximputed**). To study severe multicollinearity, which inflates the variance of estimated results and leads to unstable results, we calculated the VIF. Since several variables had $VIF > 10$, we performed an **iterative backward elimination**. By sequentially removing the four variables with highest VIF, we eliminated redundancies and obtained stable predictors for regression. At this point we can finally apply three different models, setting the hyperparameters to balance the trade off between robustness and efficiency, and compare their performances: *lm*, *lmrob* and *cellLTS*. By calculating and comparing the coefficients across the models, we noticed that classical methods are more vulnerable to contamination, showing severe **estimation bias**. For example, considering **avganncount** the OLS model underestimates its effect (due to **masking effect**) yielding a coefficient of -4.338. Both the robust methods detect this and adjust the estimate downwards: DDC MM-estimator yields -152.096 and cellLTS yields -8.737. To visually evaluate the models, we plotted the observed versus predicted values. The OLS data cloud shows dispersion around the identity line, clearly suffering from the masking effect as the model compromises its fit to accommodate extreme observations. Conversely, the MM-estimator data cloud exhibits a tighter and more concentrated fit along the diagonal, successfully resisting the pull of extreme values. The diagnostic plot for cellLTS shows higher dispersion than the MM-estimator, because of its trimming-based architecture that does not benefit data imputation.

Furthermore, the **diagnostic outlier map** (Robust Distance vs Residuals) for the robust fit highlights the most severe casewise anomalies. In a non-robust scenario they would act as **severe leverage point**, but in this case the robust model is able to successfully identify and down-weight them. Finally, we compared the **residuals scales** (MAD) across the models, obtaining 15.98 for classic OLS, 9.24 for MM-estimator and 15.62 for cellLTS. While cellLTS appears similar to OLS, this reflects a methodological difference. Indeed DDC's pre-imputation phase artificially tightens the data distribution towards the center, reducing the overall variance before the regression is fitted resulting in an artificially decreased MAD. However, cellLTS calculates the robust coefficients without modifying the underlying variance of the valid data. As a consequence, its higher MAD does not imply a worse fit, but a more realistic representation of the dataset's natural variability without the decreasing variance side effects of pre-imputed. In ad-

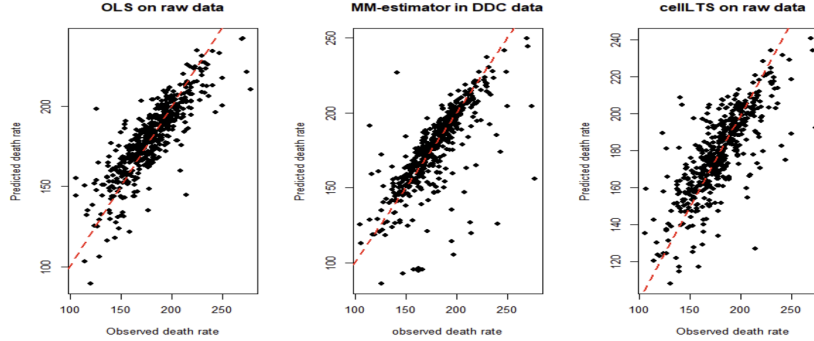


Figure 3: Observed versus predicted death rates

dition, we can also compare the R^2 value indeed also in this case the best one is the MM-estimator's one (0.8322), while both OLS and cellLTS yield values closer to 0.5. We can conclude that the difference in the behavior of the two methods is due to their distinct theoretical foundations: the DDC combined with MM-estimator maximizes global predictive accuracy by reconstructing contaminated data. Whereas cellLTS trims anomalous cells, maintaining the natural variance of the data.

References

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